

An FM modulator is followed by an ideal bandpass filter having a center frequency of 800 Hz and a Bandwidth of 120 Hz. The gain of the filter is 1 in the passband. The maximum amplitude of the carrier is 10 V & the message signal $m(t) = 5 \cos(50\pi t)$. The modulator frequency-deviation constant $k_f = 5 \text{ Hz/V}$. Determine:

- ① the peak frequency deviation in hertz.
- ② the bandwidth of the modulated signal.
- ③ the power ratio

Sol.

$$\textcircled{1} \Delta f = k_f \cdot \max. |m(t)|$$

$$\Delta f = 5 \frac{\text{Hz}}{\text{V}} \cdot 5 \text{ V} \Rightarrow \boxed{\Delta f = 25 \text{ Hz}}$$

$$\textcircled{2} B = 2 \left(1 + \frac{1}{\beta} \right) \Delta f \quad ; \quad \beta = \frac{\Delta f}{f_m} = \frac{25}{25} = 1$$

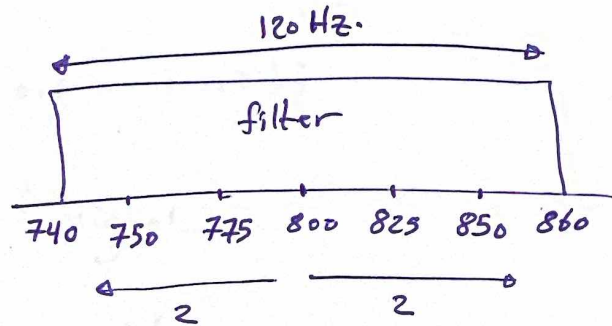
$$\therefore B = 2 \left(1 + \frac{1}{1} \right) \cdot 25$$

$$\therefore \boxed{B = 100 \text{ Hz}}$$

③

$$f_c = 800 \text{ Hz.}$$

$$f_m = 25 \text{ Hz.}$$



$$\text{power at filter input} = \frac{A_c^2}{2} = \frac{100}{2} = 50 \text{ W}$$

$$\text{power at filter o/p} = \frac{1}{2} A_c^2 \sum_{n=-K}^K J_n^2(\beta)$$

$$P_r = J_0^2(\beta) + 2 \sum_{n=1}^{\infty} J_n^2(\beta)$$

$$\therefore P_r = J_0^2(1) + 2 \sum_{n=1}^{\infty} J_n^2(1)$$

$$P_r = (0.765)^2 + 2 \cdot \left[(0.44)^2 + (0.115)^2 \right] \text{ W}$$

Ex:-

An angle-modulated signal with carrier frequency

$\omega_c = 2\pi \times 10^6$ is described by the equation :-

$$\phi_{PM}(t) = 10 \cos(\omega_c t + 0.1 \sin 2000\pi t)$$

- a) Find the power of modulated signal.
- b) Find the frequency deviation, Δf .
- c) Find the phase deviation, $\Delta \theta$.
- d) Estimate the bandwidth of $\phi_{PM}(t)$.

$$\text{a) } P = \frac{A^2}{2} = \frac{(0.1)^2}{2} = 0.005$$

$$\text{b) } f_m = \frac{2000\pi}{2\pi} = 1 \text{ KHz.}$$

$$f_i(t) = f_c + \frac{K_p}{2\pi} \cdot m'(t)$$

$$= f_c + \frac{0.1}{2\pi} [2000\pi \cdot \cos 2000\pi t]$$

$$= f_c + 100 \cos 2000\pi t$$

$$\Delta f = \max |f_i(t) - f_c| = \max |\cancel{f_c} + 100 \cos 2000\pi t - \cancel{f_c}|$$

$$\therefore \Delta f = 100 \text{ Hz.}$$

$$\textcircled{c} \quad \Delta\theta = \max |\theta(t) - 2\pi f_c t|$$

$$= \max | \cancel{2\pi f_c t} + 0.1 \sin 2000\pi t - \cancel{2\pi f_c t} |$$

$$\Delta\theta = 0.1$$

$$\textcircled{d} \quad B = 2 \left(1 + \frac{1}{\beta} \right) \Delta f$$

$$\beta = \frac{\Delta f}{f_m} = \frac{100}{1 \text{ KHz}} = \frac{100}{1000} = 0.1$$

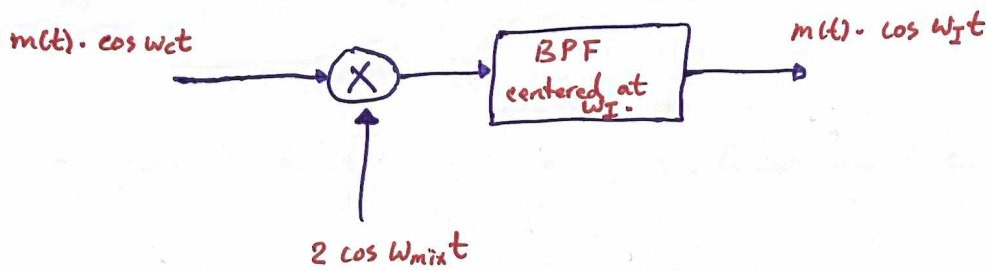
$$\therefore B = 2 \left(1 + \frac{1}{0.1} \right) \cdot 100$$

$$= 2 * 11 * 100$$

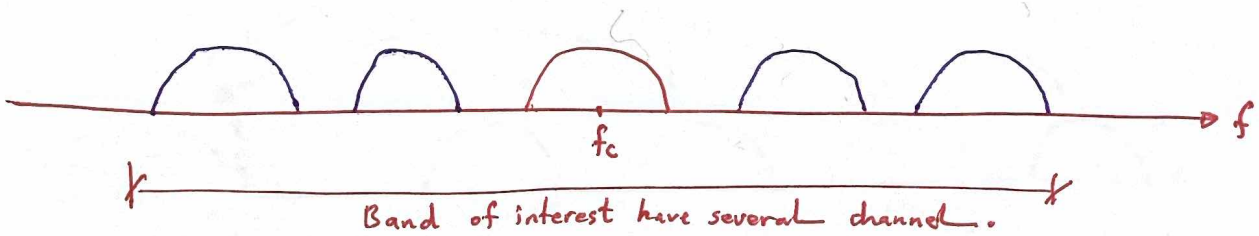
$$B = 2200 \text{ Hz}$$

$$\therefore \boxed{B = 2.2 \text{ KHz}}$$

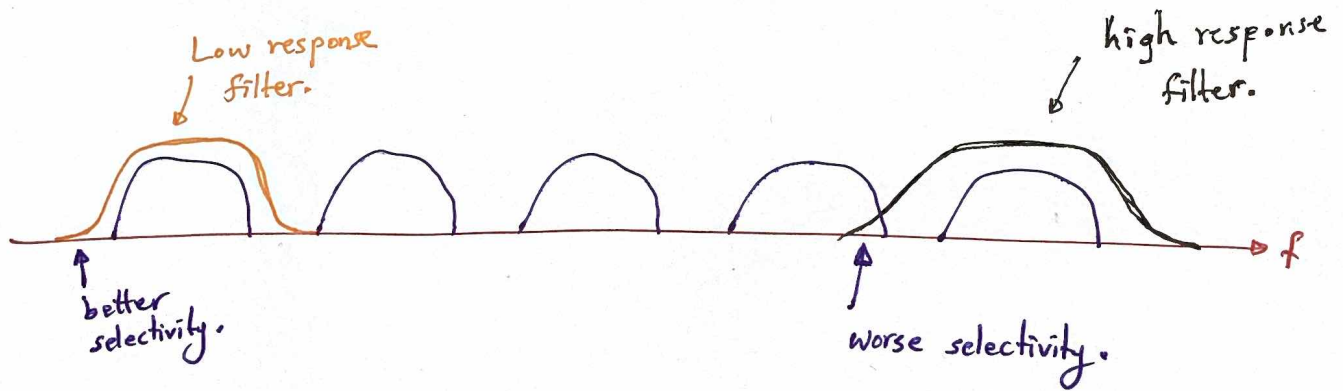
- from frequency mixer, we have ::



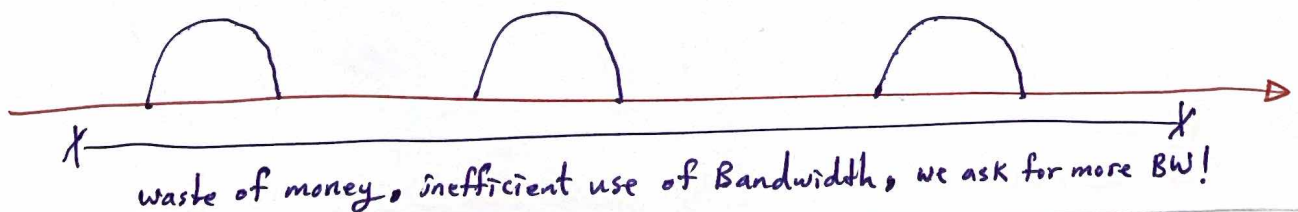
spectrum that equipment should pick up.



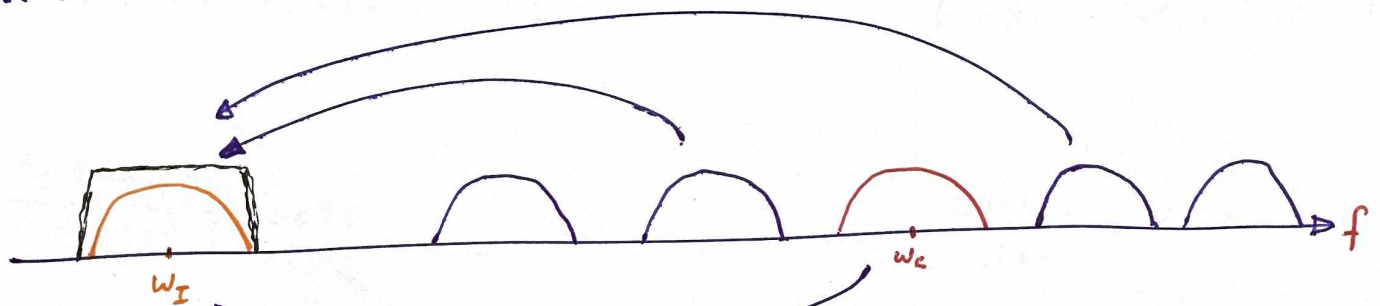
- to select any channel — ; we need a tunable filter , which is a function of f .



- Poor selectivity ; reduce the number of band ! Is this good solution? (spacing the channel).



- we should come up with another solution that still keep our Band tightly ; but still be able effectively eliminate another channel we are not interested in.
- equipment tuned for a single, lower band has better performance.
- we don't need tunable filter, just one filter.



we took the channel of interest &
move it down to an intermediate channel.

for AM Radio ; $f_I = 455 \text{ KHz}$.

Superhetrodyne

$$\omega_{mix} = \omega_c + \omega_I$$

$$530 \text{ KHz} + 455 \text{ KHz} = 1005 \text{ KHz}$$

$$1600 \text{ KHz} + 455 \text{ KHz} = 2055 \text{ KHz}$$

$$\frac{2055 \text{ KHz}}{1055 \text{ KHz}} = 2.045$$

small
tuning ratio
compared to

- small ratio means easier to tune.

Subhetrodyne.

$$\omega_{mix} = \omega_c - \omega_I$$

$$530 \text{ KHz} - 455 \text{ KHz} = 95 \text{ KHz}$$

$$1600 \text{ KHz} - 455 \text{ KHz} = 1145 \text{ KHz}$$

$$\frac{1145 \text{ KHz}}{95 \text{ KHz}} = 12.05$$

the higher
of this ratio,
the harder to
tune & get worse
result.

⊗ for TV reception; $IF = 38 \text{ MHz}$.

⊗ Based on the Federal Communication Commission (FCC) :-

AM

- allocated band (530 - 1600) KHz.
- 10 KHz per channel.
- $IF = 455 \text{ KHz}$.
- $\frac{1600 - 530}{10} = 107$ channels or station can be licensed.

FM

- The spectrum for FM broadcasting (88 - 108) MHz.
- Separation of 200 KHz between adjacent stations.
- $IF = 10.7 \text{ MHz}$.

Superhetrodyne Receiver

$$x(t) = m(t) \cdot \cos(\omega_c t) \cdot 2 \cos(\omega_{mix} t).$$

where $\omega_c = \omega_c + \Delta\omega \longrightarrow$ to move on channels within interested band.

$\omega_{mix} = \underbrace{\omega_c + \Delta\omega}_{\text{this term is changing depending on channel of interest.}} + \omega_I \longrightarrow$ (we use this b/c it give us a small tuning ratio 1:2 [1005 KHz to 2055 KHz]).

same as ω_{LO}

$$x(t) = m(t) \cdot \cos((\omega_c + \Delta\omega)t) \cdot 2 \cos[(\omega_c + \Delta\omega + \omega_I)t].$$

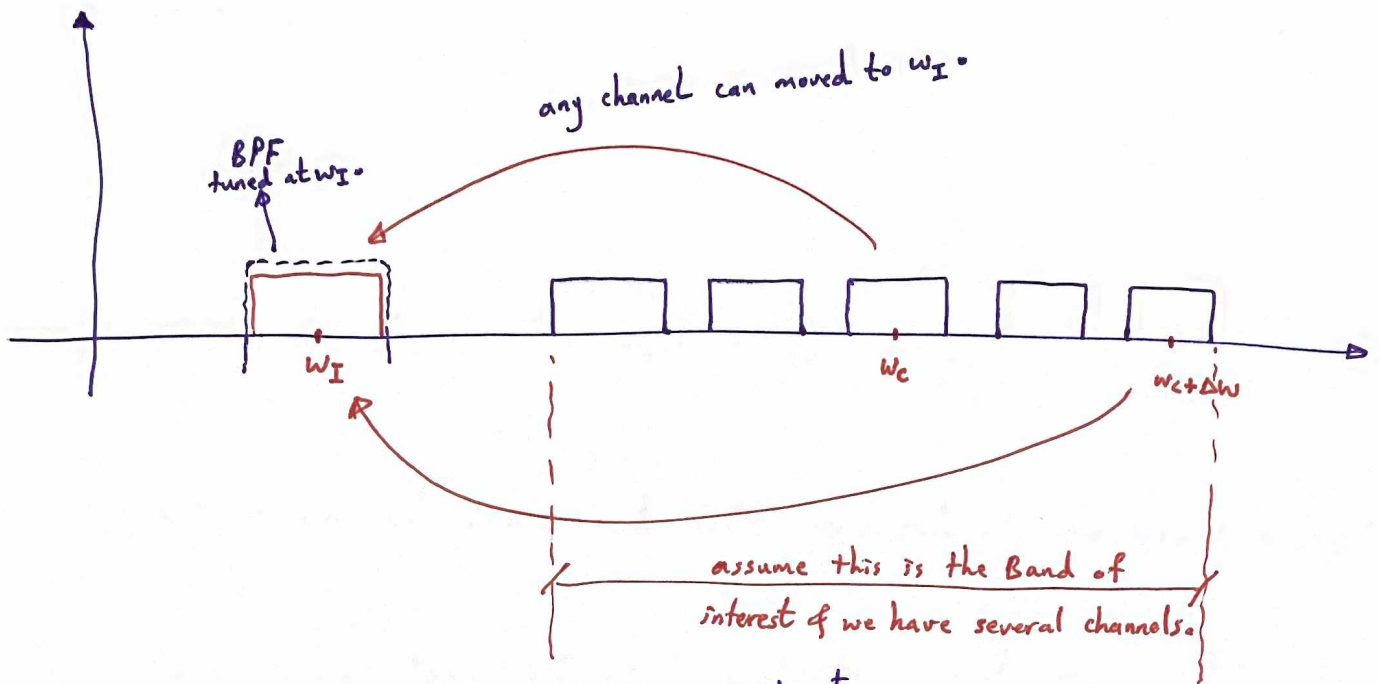
$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)]$$

$$x(t) = \cancel{2} m(t) \cdot \frac{1}{\cancel{2}} \left[\cos[(\omega_c + \Delta\omega + \omega_c + \Delta\omega + \omega_I)t] + \cos[(\cancel{\omega_c + \Delta\omega} - \cancel{\omega_c + \Delta\omega} - \omega_I)t] \right]$$

$\cos(-\theta) = \cos \theta.$

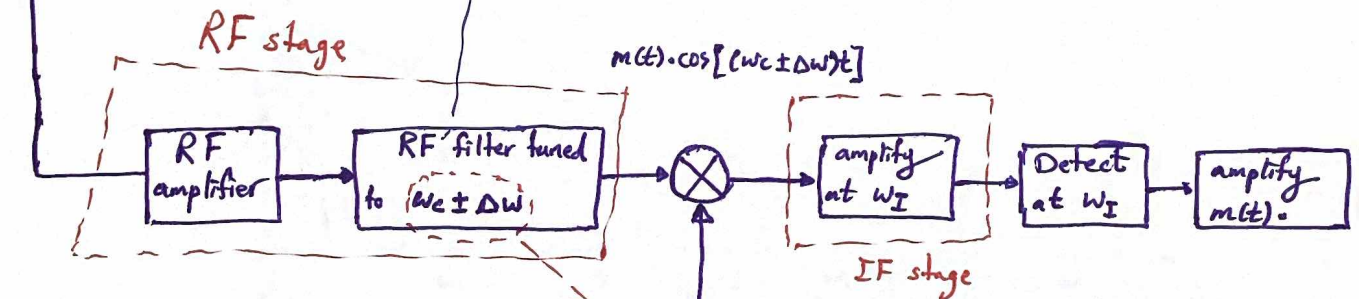
$$x(t) = m(t) \cdot [\cos[2(\omega_c + \Delta\omega)t + \omega_I t] + m(t) \cdot \cos(\omega_I t)].$$

- No matter what $\Delta\omega$ is ; we end up with two term. One is the modulated signal transformed to intermediate frequency, which is interesting one. The other can be removed by BPF tuned at ω_I .



antenna received band in some range $w_c \pm \Delta w$.

adjacent
Poor selectivity against stations separated by 10 KHz,
but reasonable selectivity against a station separated by 910 KHz ($2w_I$).



link the tuning together
"Changing the capacitor"

$$\cos[(w_c \pm \Delta w)t + w_I t]$$

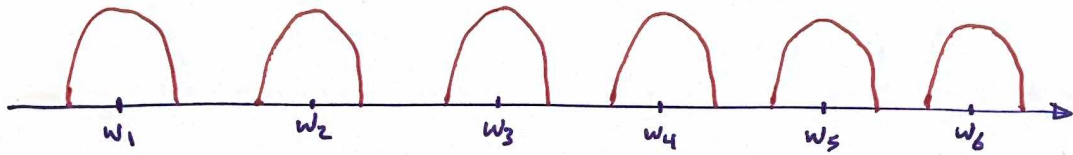
Local oscillator (LO)

constant b/c w_I is fixed.

simultaneous tuning of the local oscillator & the RF tunable filter is done by one joint knob.

this is changing depends on channel of interest.

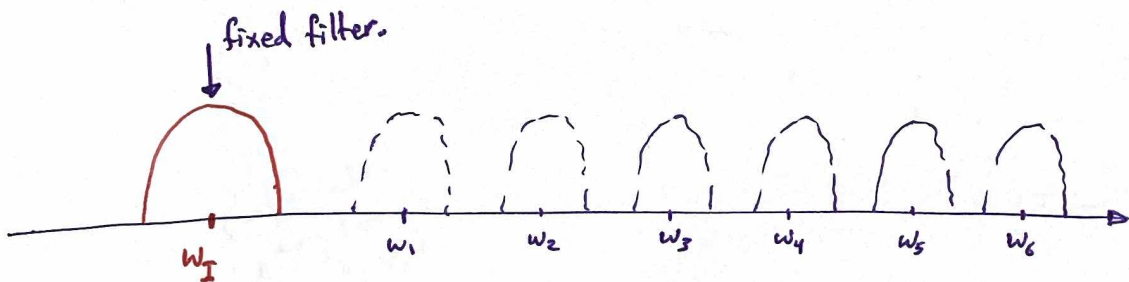
- Band with multiple channel .



- To select a channel ; we need BPF centered at w_1, w_2, w_3, w_4, w_5 & w_6

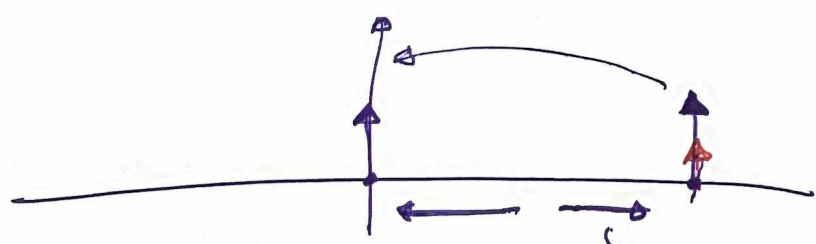
- المشكلة انه اداء الفلتر يتكلف من الترددات القليلة الى الترددات العالية فانه اداء الفلتر في الترددات القليلة يكون افضل من الترددات العالية .

- هل ممكن نعمل فلتر واحد فقط ضمن تردد معين لاستقبال جميع القنوات !!!



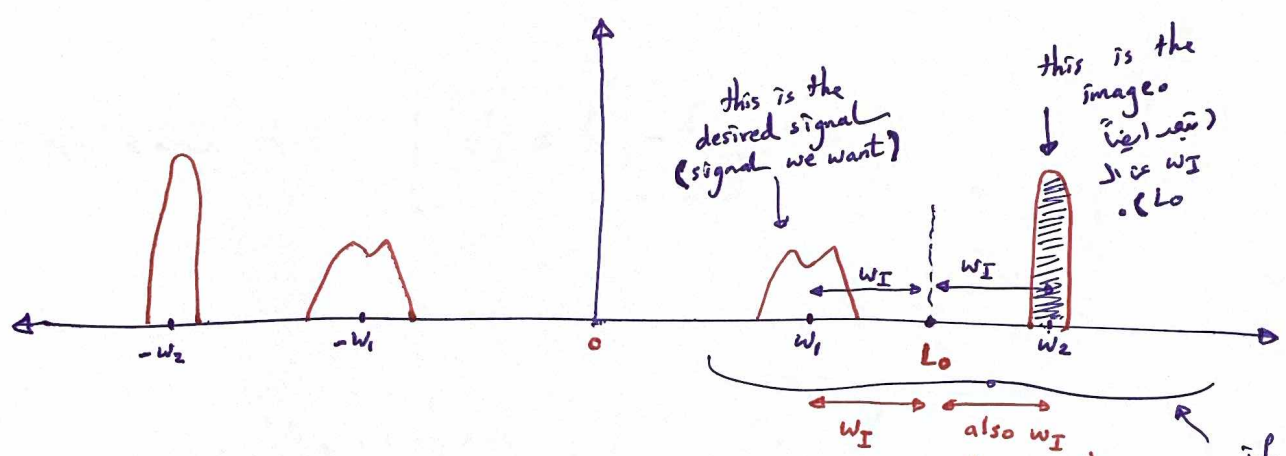
sliding to w_I .
عند اختيار اي قناة نعمل لها shift عند w_I

$L_o = \omega_c - \omega_i$
 $\omega_I = \omega_c - \omega_{mix}$



if ω_c is higher
 $\omega_{mix} = \omega_{L_o}$
 then ω_c is lower
 the higher L_o
 the lower L_o
 I move.

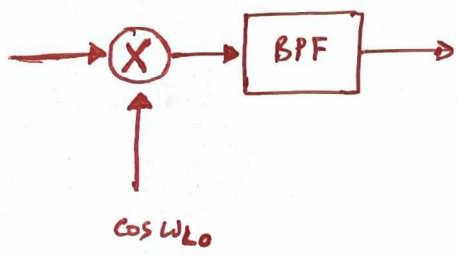
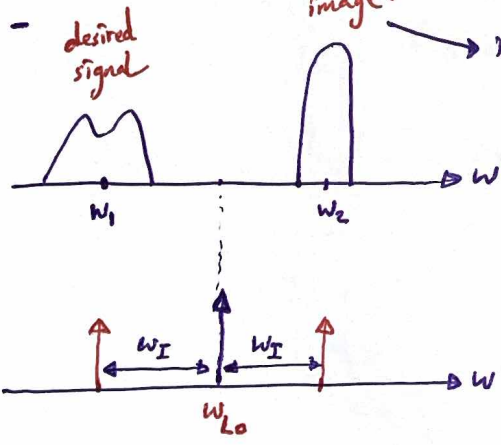
⊗ what limit us from getting my L_o close to our signal!



if we convert these two signal what will happen?

(any signal away from L_o by ω_I).
 image.

it is obvious that
 $\omega_{image} = \omega_c + 2\omega_I$



- we get the desired signal & the image at ω_I .

⊗ بحيث افترانه عننا الى desired signal
 الى L_o بمقدار ω_I فانه الجانب الاضيق الى
 L_o الذي يبعد بمقدار ω_I ايها هو ال
 image وسيتحول مع الاشارة المتكوبة
 من نفس الى ω_I لذلك يجب التخلص منه

Ex:-> AM example :-

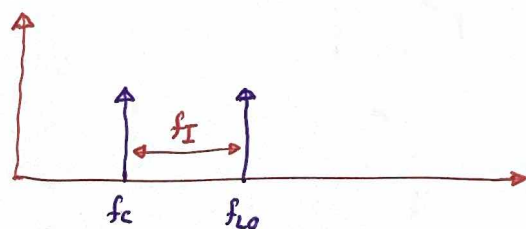
if we want to listen to station $f_c = 1000 \text{ KHz}$.

⇒ we adjust the local oscillator (LO) in the receiver to:

$$f_{LO} = f_c + f_I = 1000 \text{ KHz} + 455 \text{ KHz} = 1455 \text{ KHz}.$$

✓ we thought this f_{LO} is unique for this station $f_c = 1000 \text{ KHz}$.

we know that $f_I = f_{LO} - f_c$



Now,

for same $f_{LO} = 1455 \text{ KHz}$; check the station $f_{c_{image}} = f_{ci} = 1910 \text{ KHz}$.

we should have $f_I = 455 \text{ KHz}$.

$$f_I = f_{LO} - f_c = 1455 \text{ KHz} - 1910 \text{ KHz} = -455 \text{ KHz}.$$

remember; the o/p of frequency mixer is:

$$x(t) = m(t) \cdot \cos \omega_I t + m(t) \cdot \cos (2\omega_c + \omega_I) t.$$

$$\cos (-455 \text{ K})$$

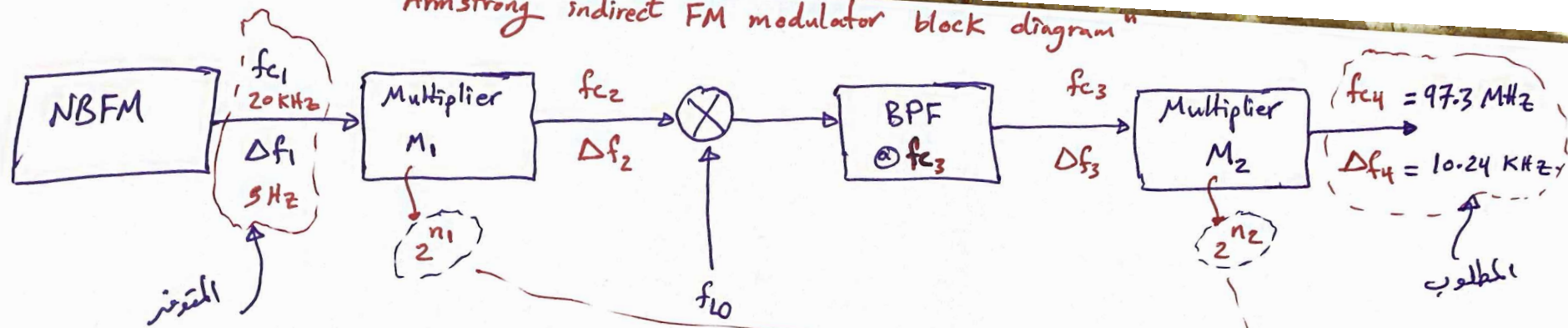
"

$$\cos (455 \text{ K}).$$

hence for $f_{LO} = 1455 \text{ KHz}$; we received two station [desired + Image].

adjacent station, removed by RF filter. image $2\omega_I$ و عليه ممكن ان يخل بتداخل مع ال adjacent station. لا يحتاج ان يكون الفلتر فيه $2\omega_I$ $\leftarrow$ removed by RF filter.

"Armstrong indirect FM modulator block diagram"



$$M_1 \cdot M_2 = \frac{\Delta f_4}{\Delta f_1} = \frac{10.24 \text{ kHz}}{5} = 2048 = 2^{11}$$

$$\Rightarrow \frac{n_1}{2} \cdot \frac{n_2}{2} = \frac{11}{2} \Rightarrow \frac{n_1 + n_2}{2} = \frac{11}{2} \Rightarrow \boxed{n_1 + n_2 = 11}$$

$$\boxed{f_{c2} = 2^{n_1} \cdot f_{c1}}$$

or

$$\boxed{f_{c4} = 2^{n_2} \cdot f_{c3}}$$

where

$$\begin{matrix} M_1 = 2^{n_1} \\ M_2 = 2^{n_2} \end{matrix}$$

b/c
it should
be
doublers as
requested.

To find f_{L0} ; we have 3 options:-

- ① $f_{c3} = f_{c2} - f_{L0}$
- ② $f_{c3} = f_{L0} - f_{c2}$
- ③ $f_{c3} = f_{L0} + f_{c2}$

- we have a constraint that $400\text{kHz} \leq f_{LO} \leq 500\text{kHz}$.

So, we test all options:-

$$\textcircled{1} \quad f_{c3} = f_{c2} - f_{LO} \Rightarrow \boxed{f_{LO} = f_{c2} - f_{c3}}$$

$$\therefore f_{LO} = \frac{n_1}{2} \cdot f_{c1} - \frac{f_{c4}}{\frac{n_2}{2}} = \frac{1}{2^{n_2}} \left[\frac{n_1}{2} \cdot 2^{n_2} f_{c1} - f_{c4} \right]$$

$$= \frac{1}{2^{n_2}} \left[2^{n_1+n_2} \cdot f_{c1} - f_{c4} \right]$$

$$= \frac{-n_2}{2} \left[M_1 \cdot M_2 \cdot f_{c1} - f_{c4} \right]$$

$$= \frac{-n_2}{2} \left[2048 * 20\text{kHz} - 97.3\text{MHz} \right]$$

$$= \frac{-n_2}{2} \left[40.96\text{MHz} - 97.3\text{MHz} \right]$$

$$= \frac{-n_2}{2} \cdot (-56.34\text{MHz}) \quad (\text{negative outside range}).$$

$$\textcircled{2} \quad f_{c3} = f_{LO} - f_{c2} \Rightarrow \boxed{f_{LO} = f_{c2} + f_{c3}}$$

$$f_{LO} = \frac{n_1}{2} \cdot f_{c1} + \frac{f_{c4}}{\frac{n_2}{2}} = \frac{1}{2^{n_2}} \left[\frac{n_1}{2} \cdot 2^{n_2} f_{c1} + f_{c4} \right] = \frac{1}{2^{n_2}} \left[2^{n_1+n_2} f_{c1} + f_{c4} \right]$$

$$= \frac{1}{2^{n_2}} \left[M_1 \cdot M_2 f_{c1} + f_{c4} \right] = \frac{-n_2}{2} \left[2048 * 20\text{kHz} + 97.3\text{MHz} \right]$$

$$= \frac{-n_2}{2} \cdot 138.26\text{MHz} = \frac{138260}{\frac{n_2}{2}} \text{kHz}.$$

for $n_2 = 8 \Rightarrow f_{LO} = 540\text{kHz}$ (above range).
 $n_2 = 9 \Rightarrow f_{LO} = 270\text{kHz}$ (below range). } not possible option.

$$\textcircled{3} \quad f_{c3} = f_{LO} + f_{c2} \Rightarrow \boxed{f_{LO} = f_{c3} - f_{c2}}$$

from point ①; we have $f_{LO} = \frac{56340}{\frac{n_2}{2}} \text{ KHz}.$

If we choose $n_2 = 7 \Rightarrow f_{LO} = 440 \text{ KHz}.$

$$\therefore \boxed{f_{LO} = 440 \text{ KHz}.$$

$$n_1 = 11 - 7 = 4$$

$$\boxed{M_1 = 2^{n_1} = 2^4 = 16}$$

$$\boxed{M_2 = 2^{n_2} = 2^7 = 128.}$$

H.W



Problems 5.3.1 & 5.3.2.